

Theoretical Physics for Mathematicians

Home Work 2

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September 24, 2026

Question 1 - ★ Hamilton's Principle: The Euler–Lagrange equations and their invariances

Let $L = L(q, \dot{q}; t)$ be smooth and let

$$S[q] = \int_{t_0}^{t_1} L(q(t), \dot{q}(t); t) dt.$$

(a) **(Fundamental lemma).** Let $F : [t_0, t_1] \rightarrow \mathbb{R}^f$ be continuous and suppose

$$\int_{t_0}^{t_1} F_a(t) \eta^a(t) dt = 0$$

for every smooth η with compact support in (t_0, t_1) . Prove that $F = 0$. *Hint: if $F_a(t_*) \neq 0$ for some t_* and some index a , construct a bump function concentrated near t_* .*

(b) Compute the first variation

$$\delta S[q](\eta) = \left. \frac{d}{d\epsilon} S[q + \epsilon \eta] \right|_{\epsilon=0}$$

for variations with $\eta(t_0) = \eta(t_1) = 0$, and deduce from (a) that $\delta S = 0$ is equivalent to the Euler–Lagrange equations.

(c) Let $F = F(q, t)$ be smooth and set

$$L' = L + \frac{d}{dt} F(q, t), \quad \text{i.e.} \quad L' = L + \frac{\partial F}{\partial q^a} \dot{q}^a + \frac{\partial F}{\partial t}.$$

Show that L and L' have exactly the same Euler–Lagrange equations. How do the actions S and S' differ?

(d) **(Coordinate invariance).** Let $q = q(Q)$ be a diffeomorphic change of generalized coordinates and let

$$\tilde{L}(Q, \dot{Q}; t) = L\left(q(Q), \frac{\partial q}{\partial Q} \dot{Q}; t\right).$$

Prove the identity

$$\frac{d}{dt} \frac{\partial \tilde{L}}{\partial \dot{Q}^b} - \frac{\partial \tilde{L}}{\partial Q^b} = \frac{\partial q^a}{\partial Q^b} \left(\frac{d}{dt} \frac{\partial L}{\partial \dot{q}^a} - \frac{\partial L}{\partial q^a} \right),$$

and conclude that the Euler–Lagrange equations are coordinate independent. Why does the Cartesian component form $m\ddot{x}^a = F^a$ acquire connection terms in curvilinear coordinates?

Answer:

Question 1 - ★ Oscillator and pendulum, with and without multipliers

(a) For

$$L = \frac{1}{2}m\dot{q}^2 - \frac{1}{2}kq^2$$

derive the equation of motion and its general solution, compute

$$E_L = \dot{q} \frac{\partial L}{\partial \dot{q}} - L$$

and verify directly that $\dot{E}_L = 0$ on solutions of the Euler–Lagrange equation.

(b) For the plane pendulum of length ℓ use the angle θ (measured from the downward vertical) to derive

$$\ddot{\theta} + \frac{g}{\ell} \sin \theta = 0.$$

Determine the small-oscillation frequency, and show that energy conservation reduces the problem to the quadrature

$$\dot{\theta}^2 = \frac{2g}{\ell} (\cos \theta - \cos \theta_0),$$

where $0 < \theta_0 < \pi$ is the amplitude of a libration. Deduce the exact period as an integral and check that it tends to $2\pi\sqrt{\ell/g}$ as $\theta_0 \rightarrow 0$.

(c) Treat the same pendulum in $\mathbb{R}^2$ with the constraint

$$G(x, y) = x^2 + y^2 - \ell^2 = 0$$

and

$$L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - mgy.$$

Write down the multiplier equations, and determine λ explicitly as a function of $(x, y, \dot{x}, \dot{y})$.

(d) Write the constraint force as

$$\vec{F}_{\text{constraint}} = -\mathcal{T} \hat{e}_r,$$

where

$$\hat{e}_r = (\sin \theta, -\cos \theta)$$

points outwards. Show that the signed inward tension is

$$\mathcal{T} = m \left(g \cos \theta + \ell \dot{\theta}^2 \right), \quad \left| \vec{F}_{\text{constraint}} \right| = |\mathcal{T}|.$$

Explain the distinction between a rigid rod, which can exert compression, and a taut string. Confirm that (c) reproduces the equation of motion of (b).

Answer: